# LEVEL UP ACADEMY

Class 10 | GSEB | Science

## Pre-Board Mock 1 (GSEB pattern)

*Chapter / Topic: Full syllabus*

Name: _____ Roll No.: _____ Date: _____

**Time: 3 hours** **Max. Marks: 80**

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**General instructions:**

1. Section A (Q1–24) is compulsory: 1 mark each.
2. Section B: answer any 9 of 12 questions (2 marks each). Section C: answer any 6 of 8 (3 marks each). Section D: answer any 5 of 7 (4 marks each).
3. Write balanced equations and draw neat, labelled diagrams wherever needed. Show formulae and units in numericals.
4. This mock follows the reported GSEB blueprint (A 24 · B 18 · C 18 · D 20). Check the latest official GSEB blueprint for changes.

### SECTION A (Q1–8): CHOOSE THE CORRECT OPTION — 8 marks

**Q1.** Which gas is given off when zinc reacts with sodium hydroxide solution? **[1]**

(a) Oxygen (b) Carbon dioxide
(c) Chlorine (d) Hydrogen

**Q2.** Which of the following is an amphoteric oxide? **[1]**

(a) $Na_2O$ (b) $Al_2O_3$
(c) MgO (d) $CO_2$

**Q3.** The functional group in ethanoic acid is: **[1]**

(a) –OH (b) –CHO
(c) –COOH (d) >C=O

**Q4.** Which component of blood helps in clotting? **[1]**

(a) Platelets (b) Red blood cells
(c) White blood cells (d) Plasma

**Q5.** Which part of the brain maintains posture and balance of the body? **[1]**

(a) Cerebrum (b) Cerebellum
(c) Medulla (d) Pons

**Q6.** The SI unit of the power of a lens is: **[1]**

(a) metre (b) watt
(c) joule (d) dioptre

**Q7.** The resistivity of a wire depends on its: **[1]**

(a) length (b) area of cross-section
(c) material (d) shape

**Q8.** In a food chain, the first trophic level is occupied by: **[1]**

(a) producers (b) herbivores
(c) carnivores (d) decomposers

### SECTION A (Q9–14): FILL IN THE BLANKS — 6 marks

**Q9.** The chemical formula of rust is _____. **[1]**

**Q10.** The pH of pure water is _____. **[1]**

**Q11.** _____ is the hardest natural substance known. **[1]**

**Q12.** The filtration unit of the kidney is called the _____. **[1]**

**Q13.** In humans, a child who inherits a _____ chromosome from the father will be a boy. **[1]**

**Q14.** 1 kWh = _____ joules. **[1]**

## SECTION A (Q15–19): STATE TRUE OR FALSE — 5 marks

**Q15.** Copper can displace iron from iron sulphate solution. **[1]**

**Q16.** Plaster of Paris should be stored in a moisture-proof container. **[1]**

**Q17.** Bile juice contains digestive enzymes. **[1]**

**Q18.** A convex mirror always forms a virtual image. **[1]**

**Q19.** Magnetic field lines intersect each other at the poles. **[1]**

## SECTION A (Q20–24): ANSWER IN ONE WORD — 5 marks

**Q20.** Name the process by which green plants make their own food. **[1]**

**Q21.** Name the gas that turns lime water milky. **[1]**

**Q22.** Which defect of vision in young people is corrected using a convex lens? **[1]**

**Q23.** Name the instrument used to measure potential difference. **[1]**

**Q24.** Which gas in the upper atmosphere protects us from ultraviolet radiation? **[1]**

## SECTION B — answer any 9 of 12 · 18 marks

*Answer any 9 questions. 2 marks each.*

**Q25.** Write balanced chemical equations, with states, for: (a) hydrogen + chlorine → hydrogen chloride, (b) barium chloride + aluminium sulphate → barium sulphate + aluminium chloride. **[2]**

**Q26.** What is a neutralisation reaction? Give two examples from daily life. **[2]**

**Q27.** What is an alloy? How is stainless steel made, and why is it useful? **[2]**

**Q28.** What is catenation? Name one element other than carbon that shows it. **[2]**

**Q29.** Give two differences between arteries and veins. **[2]**

**Q30.** Why is diffusion not enough to meet the oxygen needs of large multicellular organisms like humans? **[2]**

**Q31.** What are tropic movements? Give one example each of positive phototropism and positive geotropism. **[2]**

**Q32.** Name the male and female reproductive parts of a flower, along with their components. **[2]**

**Q33.** What is a gene? Where are genes located? **[2]**

**Q34.** Give two differences between a real image and a virtual image. **[2]**

**Q35.** Find the resistance of a bulb rated 100 W, 220 V, and the current it draws. **[2]**

**Q36.** Why should biodegradable and non-biodegradable wastes be collected in separate dustbins? **[2]**

## SECTION C — answer any 6 of 8 · 18 marks

*Answer any 6 questions. 3 marks each.*

**Q37.** What is a redox reaction? In the reaction $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$, identify the substance oxidised and the substance reduced. **[3]**

**Q38.** Write balanced equations for the reactions of: (i) sodium with cold water, (ii) iron with steam, (iii) calcium with water. **[3]**

**Q39.** How is bleaching powder prepared? Write the equation and give two uses. **[3]**

**Q40.** Give three differences between the transport of water and the transport of food in plants. **[3]**

**Q41.** Why is the pancreas called both an exocrine and an endocrine gland? **[3]**

**Q42.** Describe any three methods of asexual reproduction, with one example each. **[3]**

**Q43.** An object is placed 30 cm from a concave lens of focal length 20 cm. Find the position and nature of the image, and the magnification. **[3]**

**Q44.** What is a solenoid? Describe the magnetic field inside and outside it, and give two ways to make the field stronger. **[3]**

## SECTION D

answer any 5 of 7 · 20 marks

*Answer any 5 questions. 4 marks each.*

**Q45.** Give two chemical reactions of ethanol with equations, and two uses of ethanol. **[4]**

**Q46.** Describe the extraction of zinc from zinc blende (ZnS), with equations. Why is the sulphide ore converted into an oxide first? **[4]**

**Q47.** Describe the role of the mouth, stomach and small intestine (with the liver and pancreas) in the digestion of food. **[4]**

**Q48.** Describe the path of air in the human respiratory system and explain how we breathe in. **[4]**

**Q49.** Show sex determination in humans with the help of a cross. Explain why the chance of having a boy or a girl is equal. **[4]**

**Q50.** (a) State Ohm's law. [1]
(b) Name the instruments needed to verify it and how each is connected. [1]
(c) The potential difference across a resistor is 12 V when a current of 2.5 mA flows through it. Find its resistance. [2] **[4]**

**Q51.** (a) What is hypermetropia? Give two causes and name the correcting lens. [3]
(b) Why does the sky appear blue? [1] **[4]**

*~ All the best for your boards! ~*